

MedShield Twin

AI-Powered Smart Hospital Security Audit Report | Generated: 2026-07-09 00:19:04 by System Test Auditor

1. Executive Security Summary

This security report provides a cryptographic and analytical evaluation of the **MedShield Twin** digital hospital infrastructure. The twin currently monitors **16** active Internet of Medical Things (IoMT) devices distributed across clinical departments. Currently, **14** devices are functioning normally, **0** are flagged under active AI cyber-threat warnings, and **2** are powered down or disconnected. Cryptographic verification of historical records reports the system blockchain is in a **SECURE / VERIFIED** state.

Security Parameter	Current Value	Evaluation
Hospital Devices Tracked	16	Baseline Active Layout
Active AI Threat Warnings	0	Normal System State
Blockchain Ledger Audit	SECURE / VERIFIED	Verified Integrity

2. Departmental IoMT Device Audit

Below is the risk-assessment catalog for smart medical equipment by department:

Device ID	Name	Room / Department	Status	Risk Score
ICU-ECG-01	ECG Monitor	ICU	Online	0%
ICU-VEN-01	Smart Ventilator	ICU	Offline	0%
ICU-PMP-01	Infusion Pump	ICU	Online	0%
ER-MON-01	Patient Monitor	Emergency Room	Online	2%
ER-BPS-01	Blood Pressure Sensor	Emergency Room	Online	0%
ER-OXY-01	Oxygen Sensor	Emergency Room	Online	0%
OT-ANS-01	Anesthesia Machine	Operation Theatre	Online	0%
OT-SYS-01	Surgical Navigation System	Operation Theatre	Online	0%
RAD-MRI-01	MRI Controller	Radiology	Online	0%
RAD-XRY-01	X-Ray Modality	Radiology	Online	0%
PHR-DIS-01	Automated Drug Dispenser	Pharmacy	Online	0%
LAB-ANA-01	Blood Analyzer	Laboratory	Online	0%
GW-BED-01	Smart Bed Sensor	General Ward	Online	0%
SRV-DB-01	Hospital Core Database	Server Room	Online	0%

NET-RTR-01	Main Firewall Router	Network Control Room	Online	0%
REC-KIOSK-01	Check-in Kiosk	Reception	Offline	0%

3. AI Threat History logs

No critical cybersecurity threats or device anomalies have been flagged by the Machine Learning module in the current logging cycle.

4. Recommended Mitigation Strategies

1. Departmental Network Segmentation: Isolate clinical IoMT devices (ICU, Operation Theatre) from standard hospital WiFi networks. Apply firewalls restricting inter-departmental traffic.

2. Machine Learning Pipeline updates: Regularly update baseline dataset inputs in the *train.py* pipeline to prevent ML classification drift on newer Botnet signatures.

3. Cryptographic Ledger Backup: In case of 'TAMPERED' warnings, cross-reference the digital signatures (SIG_Admin, SIG_Analyst) stored inside the database with the secure cold backup logs.

4. Firmware Patching: Instantly isolate and patch firmware on devices indicating elevated risk scores (> 40%) to eliminate buffer overflows or protocol exploits.